

LAB EXPERIMENT RECORD

Set 18: Product Inventory Management

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	PRANAYAA S
Register Number	192524110
Date	10.10.2026
Day	Saturday

Aim

To visualize product inventory data using a bar chart, stacked bar chart, and table in R, and to build an interactive R Shiny dashboard combining all three.

Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Price (in \$)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

Note: The dataset does not include product categories, which Question 2 needs. The programs use the following categories: Product A – Electronics, Product B – Clothing, Product C – Electronics.

PROGRAM 1: Bar Chart of Product Quantities

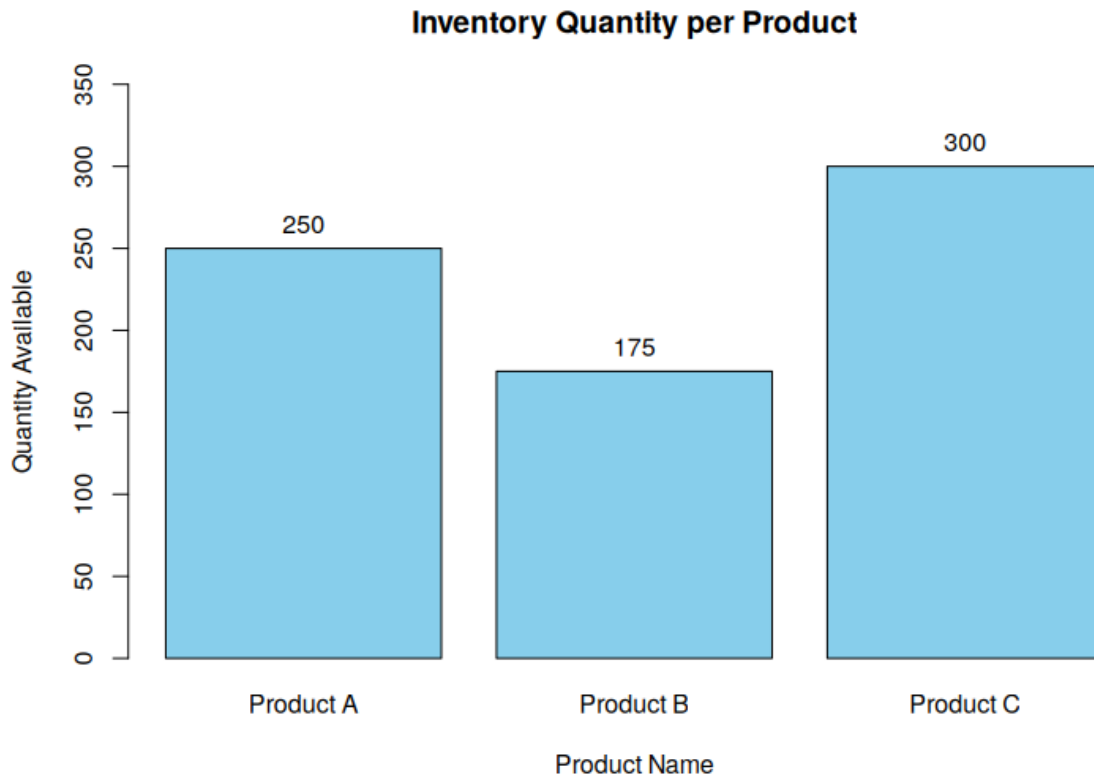
Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
# Create dataset for Set 18
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Category = c("Electronics", "Clothing", "Electronics"), # Added for Q2
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)

print(inventory_data)

# 1. Bar Chart of Quantity per Product
bp <- barplot(inventory_data$Quantity_Available,
  names.arg = inventory_data$Product_Name,
  col = "skyblue",
  xlab = "Product Name",
```

```
ylab = "Quantity Available",  
main = "Inventory Quantity per Product",  
ylim = c(0, 350))  
text(bp, inventory_data$Quantity_Available,  
      labels = inventory_data$Quantity_Available, pos = 3)
```



Output 1: Inventory Quantity per Product

Observation: Product C has the highest quantity in stock (300 units), followed by Product A (250 units). Product B has the lowest stock (175 units). The total inventory is 725 units.

PROGRAM 2: Stacked Bar Chart by Product Category

Generate a stacked bar chart to show the quantity of each product within different product categories.

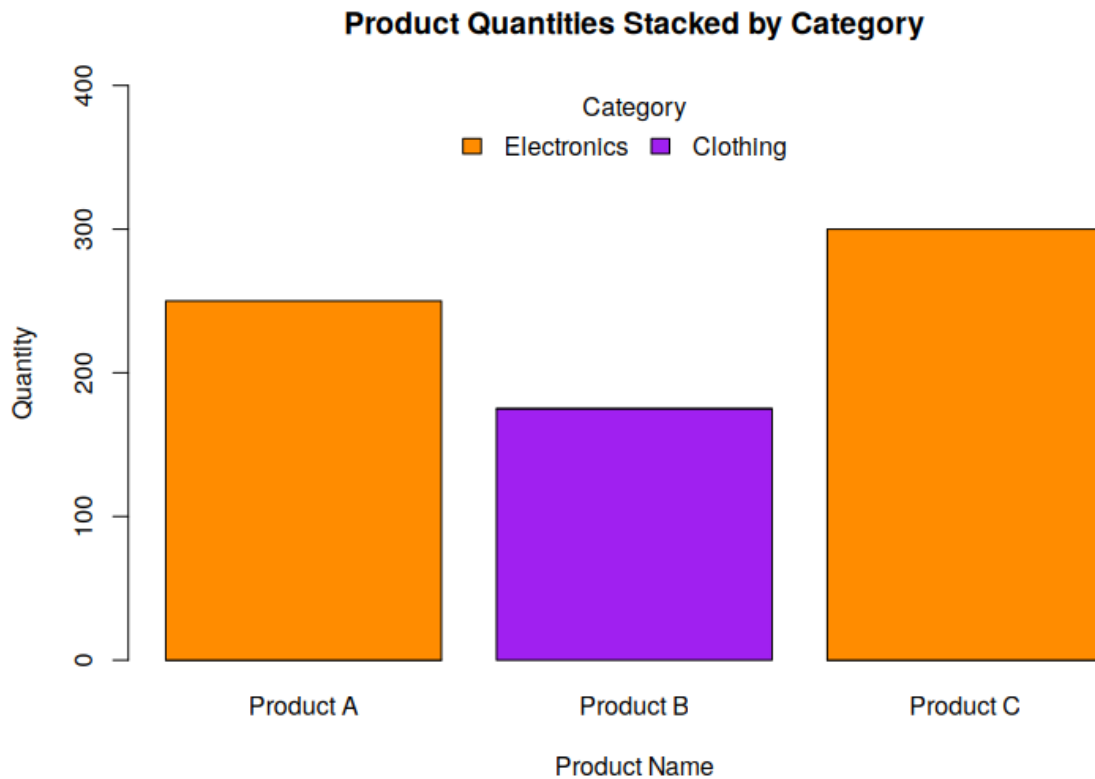
```
# Create dataset for Set 18
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Category = c("Electronics", "Clothing", "Electronics"), # Added for Q2
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)

print(inventory_data)

# Category x Product matrix of quantities
categories <- sort(unique(inventory_data$Category))
qty_matrix <- sapply(inventory_data$Product_Name, function(p)
  sapply(categories, function(c)
    sum(inventory_data$Quantity_Available[inventory_data$Product_Name == p &
      inventory_data$Category == c])))

print(qty_matrix)

# 2. Stacked Bar Chart of Quantity by Category
barplot(qty_matrix,
  col = c("purple", "darkorange"),
  xlab = "Product Name",
  ylab = "Quantity",
  main = "Product Quantities Stacked by Category",
  ylim = c(0, 400),
  legend.text = categories,
  args.legend = list(title = "Category", x = "top", horiz = TRUE, bty = "n"))
```



Output 2: Product Quantities Stacked by Category

Observation: Electronics (Products A and C) holds 550 units, which is about 76% of the total stock, while Clothing (Product B) holds 175 units. Since each product belongs to one category, every bar shows a single colour segment.

PROGRAM 3: Table of Inventory Data

Build a table to show the inventory data for each product. Label the table elements.

```
# Create dataset for Set 18
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Category = c("Electronics", "Clothing", "Electronics"), # Added for Q2
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)

print(inventory_data)

# 3. Table of inventory data
inventory_table <- inventory_data
colnames(inventory_table) <- c("Product ID", "Product Name", "Category",
                              "Quantity Available", "Price ($)")

cat("=== PRODUCT INVENTORY TABLE ===\n")
print(inventory_table, row.names = FALSE)
```

```
=== PRODUCT INVENTORY TABLE ===
> print(inventory_table, row.names = FALSE)
  Product ID Product Name    Category Quantity Available Price ($)
        1    Product A Electronics           250           20
        2    Product B   Clothing           175           15
        3    Product C Electronics           300           18
>
```

Output 3: Product Inventory Table

Observation: The table shows the category, quantity, and price of each product with clearly labelled columns. Product A is the most expensive (\$20), while Product B is both the cheapest (\$15) and the lowest in stock.

PROGRAM 4: Interactive Dashboard for Inventory Data

Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 18
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Category = c("Electronics", "Clothing", "Electronics"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)
cat_cols <- c(Clothing = "purple", Electronics = "darkorange")

ui <- fluidPage(
  titlePanel("Product Inventory Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("cat", "Category:", choices = names(cat_cols),
        selected = names(cat_cols))
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("barPlot", height = "300px")),
        column(6, plotOutput("stackPlot", height = "300px"))
      ),
      h4("Product Inventory Table"),
      tableOutput("invTable")
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({ subset(inventory_data, Category %in% input$cat) })

  output$barPlot <- renderPlot({
    d <- filtered()
    barplot(d$Quantity_Available, names.arg = d$Product_Name, col = "skyblue",
      ylim = c(0, 350), xlab = "Product Name", ylab = "Quantity Available",
      main = "Inventory Quantity per Product")
  }, width = 420, height = 300, res = 96)

  output$stackPlot <- renderPlot({
    d <- filtered()
    m <- sapply(d$Product_Name, function(p)
      sapply(names(cat_cols), function(c)
        sum(d$Quantity_Available[d$Product_Name == p & d$Category == c])))
    barplot(m, col = cat_cols, ylim = c(0, 400), xlab = "Product Name", ylab = "Quantity",
      main = "Quantities Stacked by Category", legend.text = names(cat_cols),
      args.legend = list(x = "top", horiz = TRUE, bty = "n"))
  })
}
```

```

}, width = 420, height = 300, res = 96)

output$invTable <- renderTable({
  d <- filtered()
  colnames(d) <- c("Product ID", "Product Name", "Category", "Quantity Available", "Price ($)")
  d
}, digits = 0)
}

shinyApp(ui = ui, server = server)

```

Product Inventory Dashboard



Output 4: Product Inventory Interactive Dashboard (R Shiny)

Observation: The dashboard shows the inventory bar chart, the category-wise stacked bar chart, and the inventory table together. Unticking a category in the Category filter removes its products from both charts and the table at the same time.

Procedure

1. Create the inventory data frame with Product ID, Product Name, Category, Quantity Available, and Price.
2. Use `barplot()` to show the quantity of each product and `text()` to add value labels.
3. Build a Category $\times$ Product matrix with `sapply()` and use `barplot()` to draw the stacked bar chart with a legend.
4. Rename the columns with `colnames()` and print the labelled inventory table.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts and table.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The product inventory data was successfully visualized using a bar chart, stacked bar chart, and table in R, and an interactive R Shiny dashboard was built combining all three views.

Key R Functions Used

Function	Purpose
<code>data.frame()</code>	Creates the inventory dataset.
<code>barplot()</code>	Creates the bar chart and stacked bar chart.
<code>text()</code>	Adds value labels to the bars.
<code>sapply()</code>	Builds the category-wise quantity matrix.
<code>colnames()</code> / <code>print()</code>	Labels and displays the inventory table.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>renderTable()</code>	Displays the filtered table on the dashboard.
<code>shinyApp()</code>	Runs the interactive dashboard.